Ethan Yu

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Education

Queen's University Expected May 2029
BASc in Mechatronics and Robotics Engineering — Dean's Scholar *Kingston, Ontario*

Technical Skills

Programming Languages: C, C++, Python, MATLAB
Embedded Systems: ARM Cortex-M4, AVR, RTOS, SPI, UART, I2C, Modbus
Systems & Tools: Git, CMake, Linux (Ubuntu), KiCad, LTspice, SolidWorks, Simulink

Experience

Queen's University May 2026 – Aug. 2026
Undergraduate Research Assistant *Kingston, Ontario*

- Developed embedded C firmware for a battery-powered wireless sensor node on the ATmega256RFR2 running Contiki OS, using **cooperative multitasking** for sensing, networking, and power management.
- Authored a SPI driver for a MAX31865 RTD-to-digital converter, including register-level configuration, conversion from ADC to temperature, and fault detection/clearing for a PT1000 temperature probe.
- Designed a request/response protocol over 6LoWPAN, RPL, and UDP that decodes compact codewords from a gateway to selectively collect and transmit only the requested sensor data.
- Implemented **deep-sleep power management** using AVR power-save mode and a wake timer to minimize energy use between transmissions, extending estimated battery life by over **17×**.

Queen's Knights Robotics Team Sept. 2026 – Present
Embedded Systems Developer *Kingston, Ontario*

- Implemented a **C++ driver** for the INA260 current sensor, handling low-level I2C register communication and sensor configuration within the robot's embedded framework.
- Developing chassis power management logic using INA260 current feedback to limit drive power when the energy buffer is not fully charged.

Projects

Modbus RTU Accelerometer Sensor Aug. 2026 – Sept. 2026

- Engineered embedded C firmware for a SAMG53-based sensor node that acquires and analyzes accelerometer data, transmitting processed measurements via **Modbus RTU** over RS485.
- Developed a SPI driver for a KX132-1211 3-axis accelerometer, using direct register-level PDC (DMA) control on the SPI peripheral to offload per-sample data transfer from the CPU.
- Implemented an RTC-driven acquisition scheduler collecting **9,000 samples** (3,000/axis) every 2 minutes, decoupling interrupt timing from blocking acquisition to prevent peripheral stalls.

Autonomous Rover Jan. 2026 – Apr. 2026

- Architected **closed-loop dual-motor speed control** in Arduino firmware, using encoder interrupts and PID controllers to regulate wheel velocity and motor PWM at 100 Hz.
- Programmed **Python ROS 2 nodes** on a Raspberry Pi to interface with Arduino over a custom 115200-baud serial protocol, sending wheel-velocity commands and receiving encoder counts for navigation.
- Configured a 500 ms command timeout that resets wheel-speed targets and PID integrators when commands stop arriving, improving robustness to communication interruptions.

Passive Wireless Temperature Sensor Aug. 2026 – Oct. 2026

- Designed and experimentally characterized multiple RLC PCB configurations from 24–160°C, measuring resonance bandwidth and S11 to evaluate temperature sensitivity.
- Built a Python model of a parallel RLC tank inductively coupled to a reading coil, computing complex input impedance and S11 from circuit parameters to predict VNA reflection measurements.
- Resolved a 10× discrepancy between predicted and measured S11 by correcting the analytical model, then validated its resonance solution against numerical results.